

Microscopy in Paleoethnobotany



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In paleoethnobotany, different types of microscopy are used to identify different types of botanical remains. Test your knowledge of macrobotanical and microbotanical remains by matching each type of remains with the type (or types) of microscopy appropriate for their analysis. You may want to first review the different types of paleoethnobotanical remains that were covered in an earlier module.

A Case Study of Microscopy: Starch Spherulites

Before continuing, make sure you read the assigned article by Emily Johnson and John Marston, considering the reading questions provided earlier and repeated below:

- What was the primary question asked by the authors that they answered through the research presented in this article?
- What is a spherulite, and what are the characteristics of starch spherulites?
- What evidence provided in the article convinced the authors that nixtamalization creates starch spherulites?

This case study begins as the story of Emily Johnson. Emily was an undergraduate student at BU, an Archaeology major who graduated in 2017. Emily is now a Ph.D. student (<https://www.anth.ucsb.edu/people/emily-johnson>) in the Department of Anthropology at the University of California, Santa Barbara. But this story begins when Emily was a junior at BU.





Emily, conducting flotation as part of the Kaymakçı Archaeological Project in 2016.

Source: Dr. Marston

Emily had been volunteering in my laboratory for a few years, focusing in paleoethnobotanical analysis of seeds using reflected light microscopy. She received UROP funding to help with two different research projects in the lab, analyzing seed remains from the sites of Kaymakçı and Gordion, in modern Turkey. But her core interest was Maya archaeology and paleoethnobotany. Emily proposed to me a senior honors thesis on the process of nixtamalization: soaking maize (corn) in lime-rich, alkaline water to transform the kernels into hominy corn, which can be ground into masa flour, from which tortillas and tamales are made. This process has never been identified archaeologically because it involves grinding grains into powder, so they would not be preserved in an identifiable fashion. But Emily reasoned that if nixtamalization affects maize starch, then starch granules might be preserved archaeologically. And we know from Maya iconography that tamales were an important part of the ancient Maya diet, as they are part of the Latin American diet today.



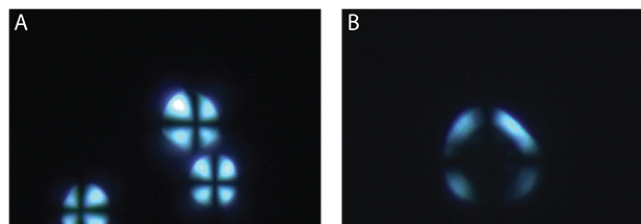
From a mural at the Maya site of San Bartolo (in modern Guatemala), a Maya woman offering a bowl of tamales to a god (who appears farther off to the right, not in this view). If you're curious how we know that these objects represent tamales, you can read this [well-illustrated article](https://www.mesoweb.com/publications/Workshop1/Taube_Works_v1.s.pdf) (https://www.mesoweb.com/publications/Workshop1/Taube_Works_v1.s.pdf) that explains the reasoning. This is the so-called "notched-ball" type of tamale.

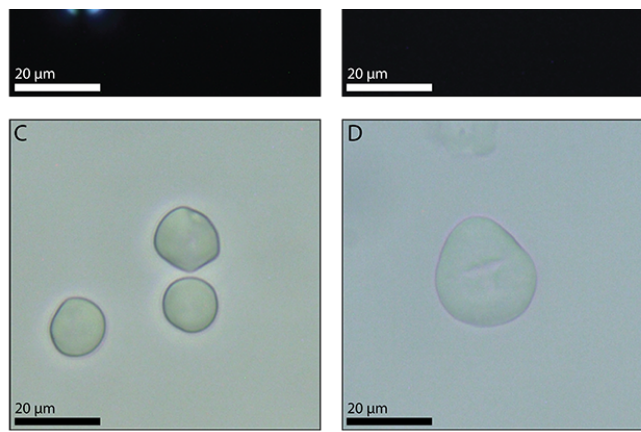
As there was no published experimental research on this question, Emily set up simple experiments. Her plan was to use traditional methods of nixtamalization likely used by the ancient Maya on heirloom maize kernels, then collect the resulting starch, and look at it under polarized light microscopy. She set up experiments in which maize was placed in one of four conditions:

	Heat	No Heat
Lime	Nixtamalized	Soaked with lime
No Lime	Boiled	Soaked

Experimental conditions, with additional variations in length of time boiled, established by Emily Johnson (see Johnson and Marston 2020, Table 1 for full details).

Emily then sampled the cooking water (termed *nejayote*, a word from Nahuatl, the language of the Aztecs), as well as the nixtamal (the cooked grains, also from a Nahuatl word; today, most often called hominy in the U.S., after its Powhatan name). Each sample was mounted on a glass slide and examined using polarized light microscopy. Emily's control samples produced what was expected. Normal starch, shown below, was present when no lime or heat was added. Heating without lime produced boiled starch types, as shown below: swollen granules with damage to the extinction cross.

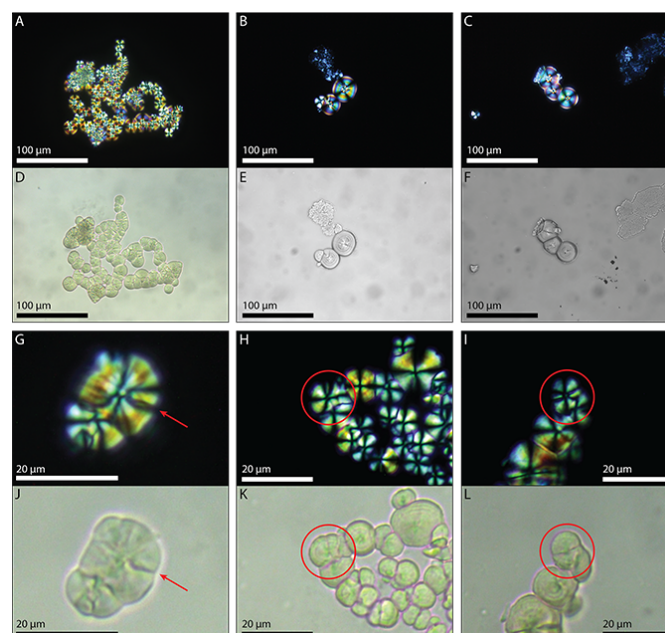




Unaltered maize starch (left pair) and boiled maize starch (right pair); top images use cross-polarized light, bottom images use normal transmitted light. Note the damaged extinction cross at top right.

Source: Johnson and Marston 2020 Figure 1

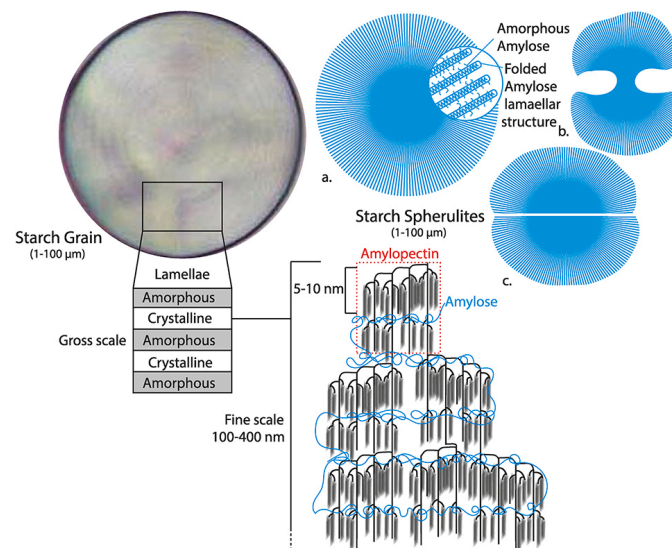
But in the nixtamalized grains and the nejayote, Emily found something surprising! Inside the nixtamalized kernels, the normal starch was present, apparently because the alkaline solution did not fully penetrate the maize kernels. But the surfaces of those kernels and the nejayote were filled with another type of starch that did not match any published information we initially could find. Emily referred to this characteristic change as the “rainbow morphology.” She also identified fissures, or cracks, in the nixtamalized starch, which appeared in cross-polarized light as extra “arms” of the extinction cross. They appear rounder, but also less smooth on the outside, and come in a wide variety of sizes, which again do not match maize starch granules that have either been soaked or boiled. Surprisingly, maize soaked in lime, without heat, also produced these unusual nixtamalized morphologies, though heating led to more starch being present in the nejayote liquid.



Starch from nixtamalized samples, displaying the rainbow nixtamalized morphology observed under cross-polarized light (A-C, G-I) and the same images in brightfield illumination (D-F, J-L). Images J and K show radial “fissures,” as observed in brightfield illumination, which appear as additional “arms” of the extinction cross (though they are not a birefringence feature) in G and H. Image L shows a fissure that transects the entire particle, and image I, the apparent refringence line corresponding to that feature.

Source: Johnson and Marston 2020, Figure 2

It took over a year, but we were finally able to understand what had happened to the maize starch that made these appear so different from normal maize-starch granules. We hypothesized that in the presence of the alkaline solution, the starch granules lost their original structure entirely, unraveling into individual starch molecules, which then spontaneously reformed into artificial structures known as spherulites. Starch spherulites are known from food science, where they have been produced under different experimental conditions. An explanation for how they form is given in another recent article by Monica Ramsey and Dano Nadel.



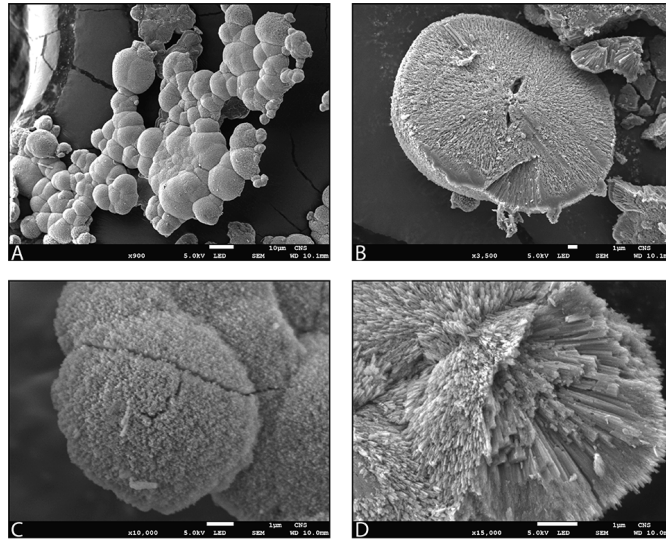
An illustration of the formation of starch granules (left and bottom) versus starch spherulites (a, b, and c). Notice that spherulites often form in the “dumbbell” shape, which leaves a line, or apparent fissure, across the spherulite.

Source: [Ramsey and Nadel](#)

(<https://www.sciencedirect.com/science/article/abs/pii/S03054403>

21001357?via%3Dihub) Figure 2

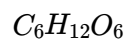
To test whether the nixtamalized starch shared other known characteristics of spherulites, we examined them using SEM-EDS at the Harvard University Center for Nanoscale Systems (<https://cns1.rc.fas.harvard.edu/>). We found that several characteristics of spherulites were even more visible at high magnification made possible by SEM: They are clearly radial balls of crystalline structures (compare D below with A above) and display a fissure across the center, where they grew in a “dumbbell” shape (compare B below with C above).



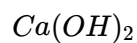
Nixtamalized starch particles observed under scanning electron microscopy. Entire structures (A), and their surface texture (C), are revealed as radial bundles of linear crystalline structures at higher magnifications (D) and when particles are broken open (B).

Source: Johnson and Marston 2020, Figure 5

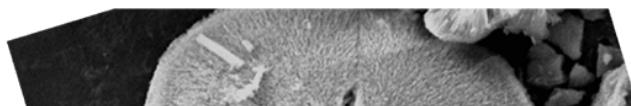
The final question was what these starch particles were made of. Starch contains the elements carbon, hydrogen, and oxygen, since it is composed of repeating units of glucose (

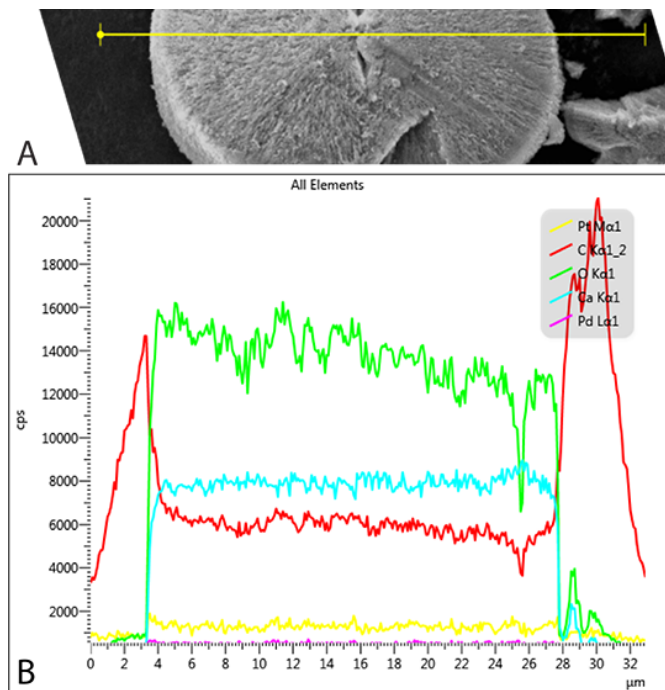


). The lime (



) added calcium into the solution. We measured the elements present across the interior surface of one of the particles and found that carbon, oxygen, and calcium were all present (hydrogen cannot be measured by EDS systems; it is too small). This meant that the spherulites continued to include calcium ions from the lime added to the nejayote solution.





SEM-EDS image (A) and spectrum (B) of a nixtamalized starch particle. The low levels of platinum (Pt) and palladium (Pd) are present due to sample coating for SEM imaging. Carbon spikes below 4 μm and beyond 28 μm are measurements of the carbon tape behind the sample, which adheres it to the stub used to transport the sample.

Source: Johnson and Marston 2020, Figure 6

This article was published in 2020 and had an immediate impact. Another pair of scholars (Ramsey and Nadel) found starch spherulites on 23,000-year-old grinding stones (<https://www.sciencedirect.com/science/article/abs/pii/S0305440321001357?via%3Dihub>) at a pre-agricultural site in Israel, which is especially exciting because it means that methods other than nixtamalization can produce starch spherulites. I worked with another research team (https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3948399) to identify starch spherulites, almost certainly from nixtamalized maize, at the Maya archaeological site of San Bartolo, Guatemala, in two large pits cut into the floors of houses. We hypothesize that these spherulites were introduced to these pits in nejayote which was used to flush toilets! This makes these both the first direct evidence of nixtamalization and the first evidence of flush toilets in the Maya world. Read more about it in BU Today (<https://www.bu.edu/articles/2022/what-ancient-toilets-can-teach-us-about-maya-life-and-tamales/>).

From this case study, including a detailed reading of the Johnson and Marston article itself, you should be able to answer these questions in detail:

1. What was the archaeological problem that interested Emily?
2. How did she design an experimental protocol to explore her research questions?
3. What features were unusual about the nixtamalized starch particles, and how were different

types of microscopy used to identify those features?

4. What evidence supported the identification of these particles as starch spherulites?